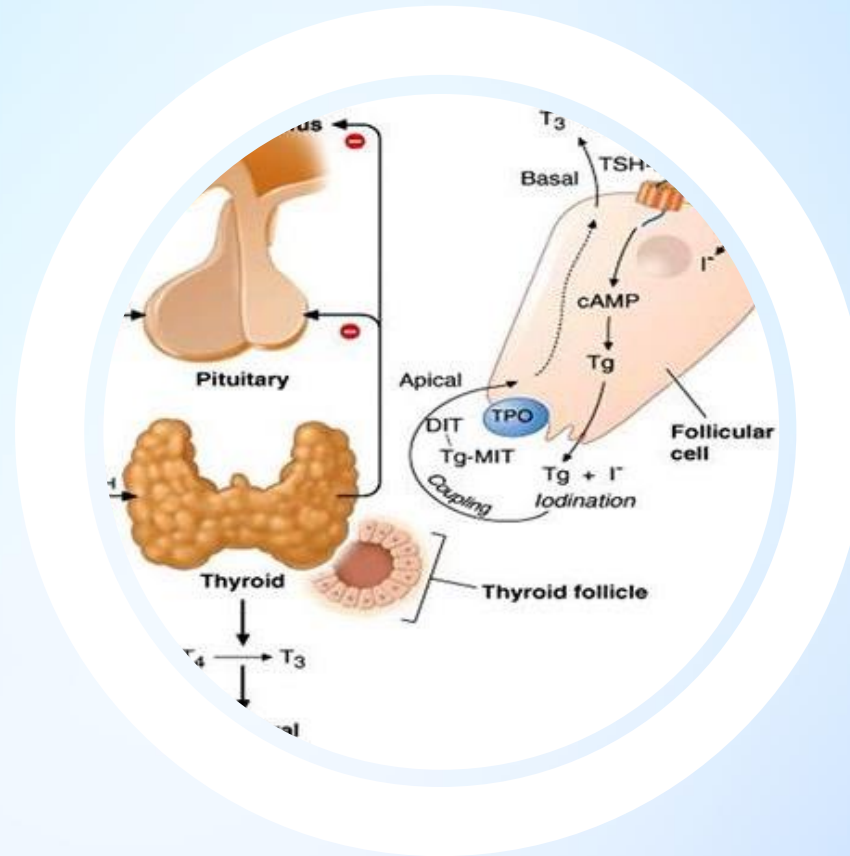




Pharmacology of thyroid and anti-thyroid drugs

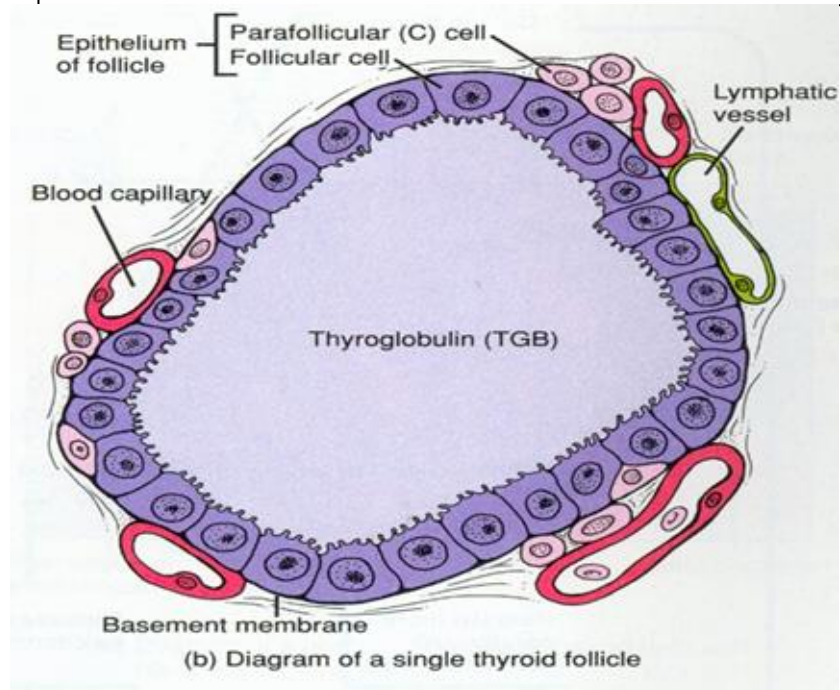
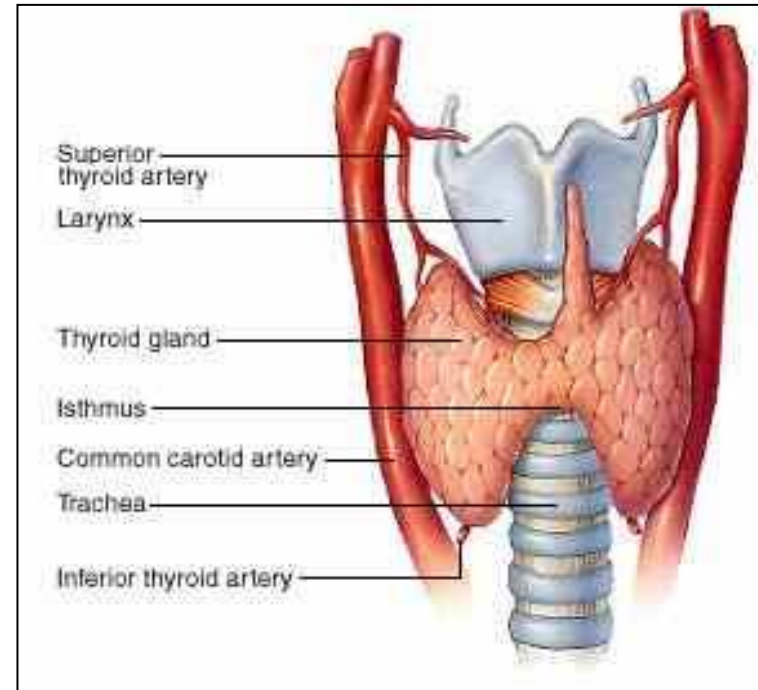


Assoc. Prof. Vilasinee Hirunpanich Sato
Department of Pharmacology Faculty of Pharmacy,
Mahidol University



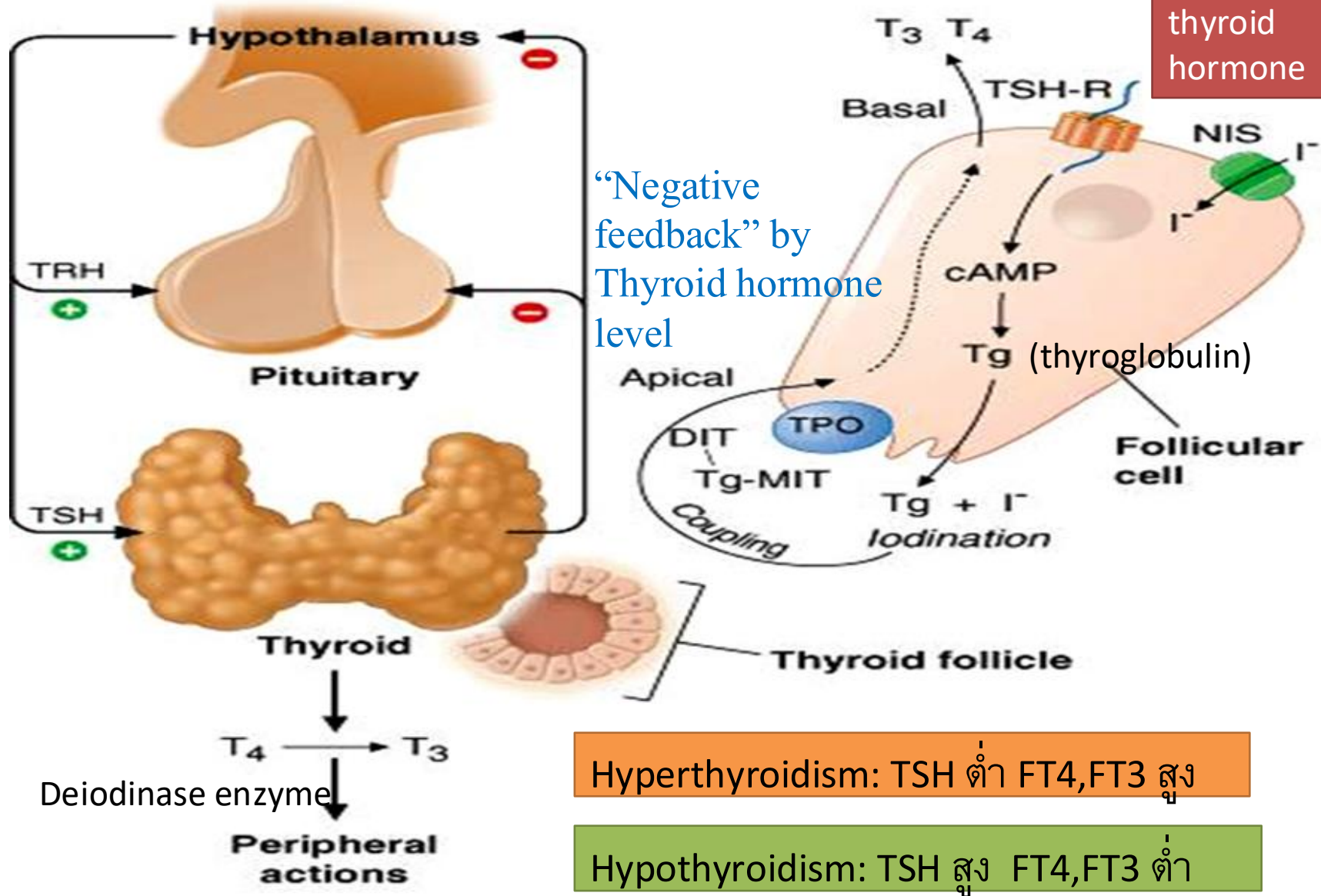
1. Thyroid gland

- **Thyroid follicles secrete thyroid hormones**
 - T4 (tetra-iodothyronine or **thyroxine**)
 - T3 (tri-iodothyronine)
- **Contained “parafollicular cells” (C-cells) produce “calcitonin hormone”**
- **Thyroid follicular colloid stores thyroid hormone as amino residues of “Thyroglobulin: TG”**



Thyroid Regulation

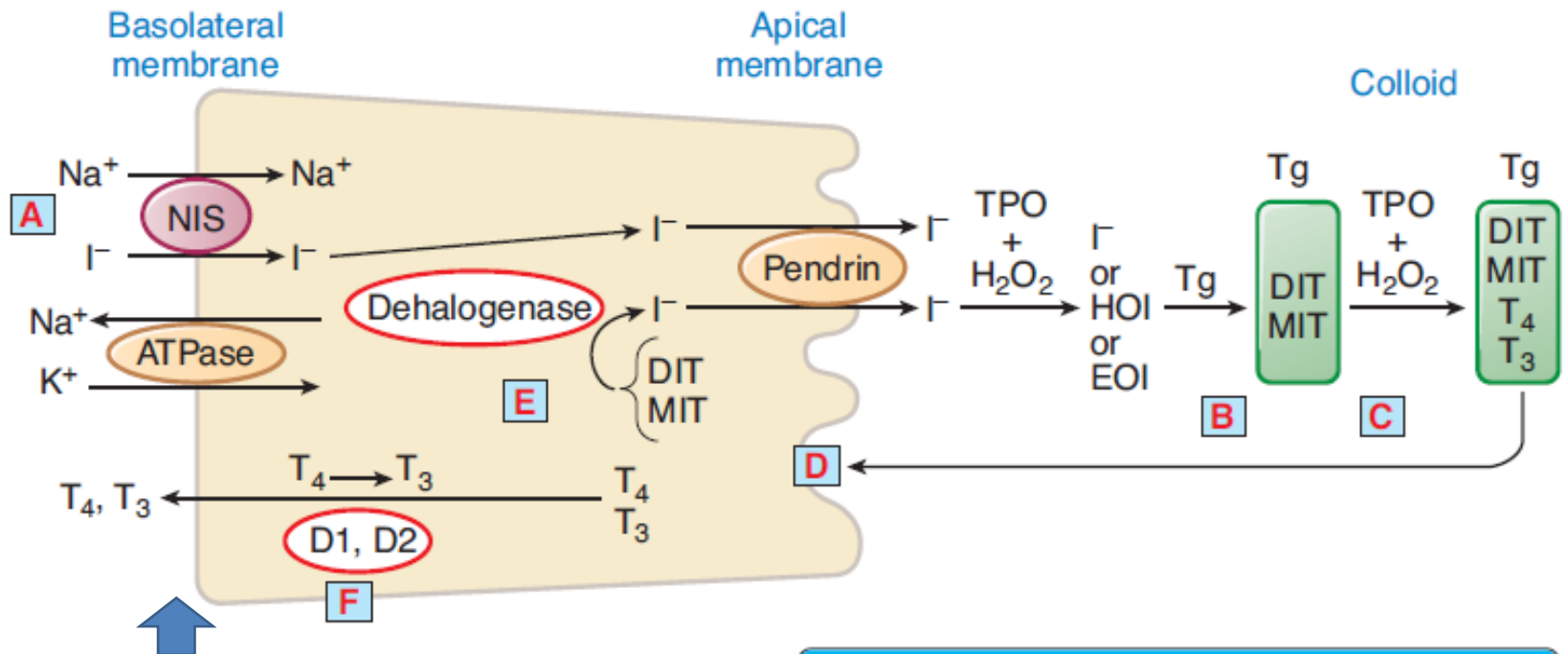
“Iodide”:
source of
synthesis of
thyroid
hormone



Steps of Thyroid Hormone Biosynthesis

1. Uptake of Iodide ion by thyroid gland (NIS: Na^+/I^- symporter)
2. Oxidation of iodide and iodination of tyrosyl groups of thyroglobulin (TG)
3. Coupling of iodotyrosine residues by ether linkage to generate the iodothyronines
 - $\text{MIT} + \text{DIT} \rightarrow \text{T3}$ (Tri-iodothyronine)
 - $\text{DIT} + \text{DIT} \rightarrow \text{T4}$ (Thyroxine)
 - rates of synthetic activity depend on the concentration of TSH and the availability of iodide.
4. Resorption of the thyroglobulin colloid from the lumen into the cell
5. Proteolysis of thyroglobulin and release of T4 and T3 into the blood
6. Recycling of the iodine within the thyroid cell via de-iodination of mono- and diiodotyrosines and reuse of the I^-
7. Conversion of thyroxine (T4) to T3 in peripheral tissue and in the thyroid

Biosynthesis of thyroid hormone



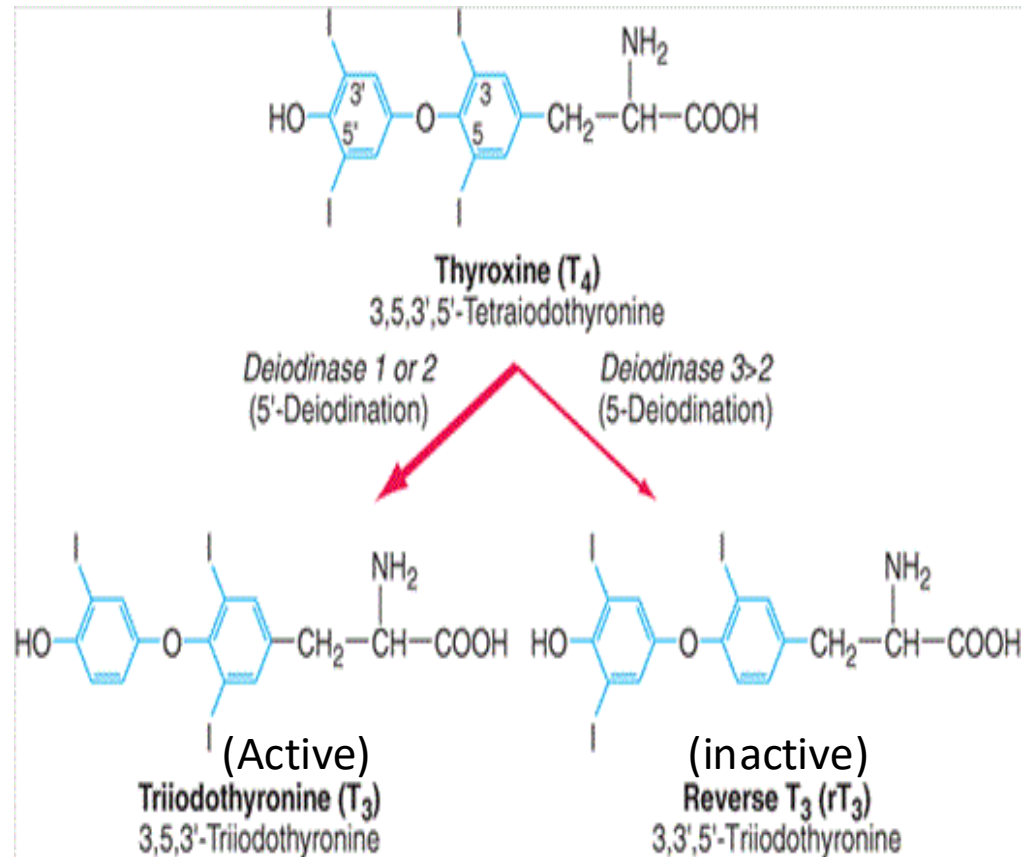
KEY

	METABOLIC STEP	INHIBITOR
A	Iodine transport	ClO_4^- , SCN^-
B	Iodination	PTU, MMI
C	Coupling	PTU, MMI
D	Colloid Resorption	Colchicine, Li^+ , I^-
E	Deiodination of $\text{DIT} + \text{MIT}$	Dinitrotyrosine
F	Deiodination of T_4	PTU

Thyroid hormone synthesis (cont.)

● Peripheral conversion from T4 to T3

- daily production of thyroxine :80-100 μg
- T3: 30 - 40 μg
- Peripheral tissues – liver and kidney
- Change T4 $\rightarrow$ T3
- >80% of T3 came from deiodination (metabolism) from T4
- T4 in plasma: 4.5-11 $\mu\text{g/dL}$
- T3: $\sim 1/100$ of that (60-180 ng/dL)



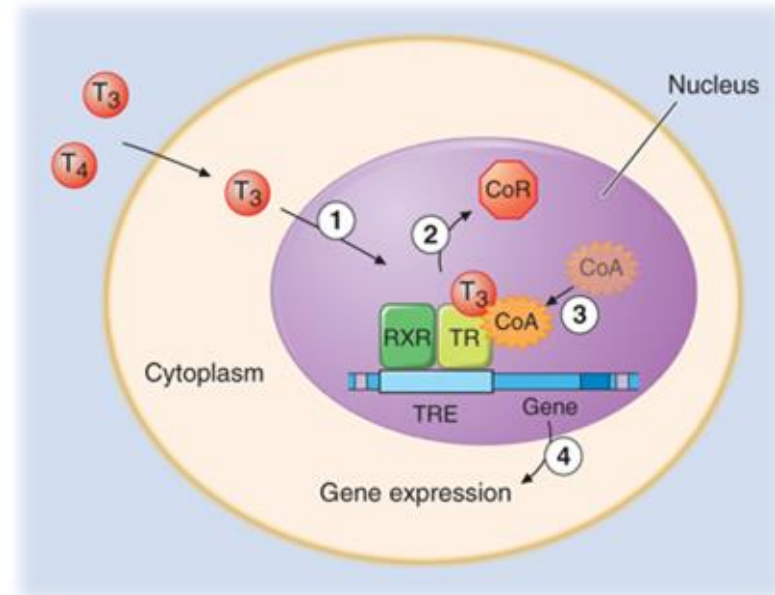
THYROID HORMONE ACTION

- **Thyroid Hormone Transport:**

- enter cells by passive diffusion and via specific transporters
 - monocarboxylate 8 transporter (MCT8), MCT10, OATP 1C1

- **Nuclear Thyroid Hormone Receptors**

- bind with high affinity to nuclear thyroid hormone receptors (TRs) α and β
- TR- α : brain, kidneys, gonads, muscle, heart
- TR- β : pituitary and liver



Thyroid abnormality

Hyperthyroidism	Hypothyroidism
An overactive thyroid resulting in overproduction of thyroid hormones	An underactive thyroid resulting in underproduction of thyroid hormones
<u>Causes</u> • Graves disease • Thyroiditis • Noncancerous growths of the thyroid gland (i.e., goiters or nodules) or pituitary gland	<u>Causes</u> • Inflammation of the thyroid gland (i.e., Hashimoto's thyroiditis) • Surgical removal of a portion or all of the thyroid gland • Radioiodine damage to the thyroid gland
<u>Symptoms</u> • Difficulty concentrating • Fatigue • Frequent bowel movements • Goiter (visibly enlarged thyroid gland) or thyroid nodules • Heat intolerance (feeling hot all of the time) • Increased appetite • Increased sweating • Irregular menstrual periods in women • Nervousness • Restlessness • Weight loss	<u>Symptoms</u> • Fatigue • Constipation • Unexplained weight gain • Puffiness or swelling in the face • Muscle aches and weakness • Depression • Cold intolerance (feeling cold all of the time) • Dry skin and hair • Hair Loss • Heavy menstrual periods • Brittle hair and nails
Diagnosis - Blood tests (TSH, free T4 and free T3), symptoms and physical exam help to establish the diagnosis	Diagnosis - Blood tests (TSH, free T4 and free T3), symptoms and physical exam help to establish the diagnosis

Treatment of hypothyroidism

- **Thyroid hormone replacement therapy: T4**
- “T4”: L-thyroxine (Levothyroxine)
 - Absorption: slightly increase in empty stomach, less variability on the TSH
 - $T_{1/2}$: 6-8 days
 - Prolong duration of action
 - Not use T3 because short $T_{1/2}$ (0.75-1 days), higher cost, transient elevation of T3 concentration above the normal range
 - Adverse effects : risk of atrial fibrillation, osteoporosis, other symptoms similar to hyperthyroidism

Drugs/dosage form	content	dose	Selected adverse events	Comment/equivalent
Thyroid USP Armour thyroid, Nature-thyroid and Westhroid (T4:T3= 4.2:1)	Desiccated pork thyroid gland	Start: 60 mcg/d Maintenance: 60-120 mcg/d	Only with excessive dosing: arrhythmia, palpitation, nervousness, tremor, weight loss	High T3:T4 ratio: inexpensive Not use much because of variability of T3 content
<u>Levothyroxine (L-thyroxine, thyroxine)</u>	Synthetic T4	Start: 12.5-100 mcg/d Maintenance: 75-150 mcg/d	Only with excessive dosing	Stable, predictable potency, generic maybe equivalent, half life 7 days <u>“consider to be drug of choice”</u>
Liothyronine	Synthetic T3	Start: 25 mcg/d Maintenance: 25-75 mcg/d	Only with excessive dosing “ cardiotoxicity ”	Uniform absorption, rapid onset, short half life 1.5 days , rapid peak and troughs ใช้แทน T4 ระยะ สั้นเท่านั้น
<u>Liotrix</u> (levothyroxine +liothyronine)	Synthetic T4:T3 ratio: 4:1	Start: levothyroxine 50 mcg/d+ liothyronine 12.5 mcg/d Maintenance: levothyroxine 100 mcg/d+liothyronine 25 mcg/d	Only with excessive dosing	Stable, predictable, expensive

Factors influencing oral levothyroxine therapy

Impaired levothyroxine absorption

Aluminum-containing antacids
Bile acid sequestrants (cholestyramine, colestipol, colesevelam)
Calcium carbonate (effect generally small)
Chromium picolinate
Food
Iron salts
Lactose intolerance (single case report)
Phosphate binders (lanthanum carbonate, sevelamer)
Proton pump inhibitors
Raloxifene
Soy products (effect generally very small)
Sucralfate

Impaired absorption

Drugs and other factors that may increase levothyroxine dosage requirements

Increased thyroxine metabolism, CYP3A4 induction of hepatic

Bexarotene
Carbamazepine
Phenytoin
Rifampin
Sertraline

CYP3A4-inducer

Drugs and other factors that may decrease levothyroxine dosage requirements

Advancing age (>65 years)

Androgen therapy in women

Drugs that may decrease TSH without changing free T_4 in levothyroxine-treated patients

Metformin

Impaired $T_4 \rightarrow T_3$ conversion

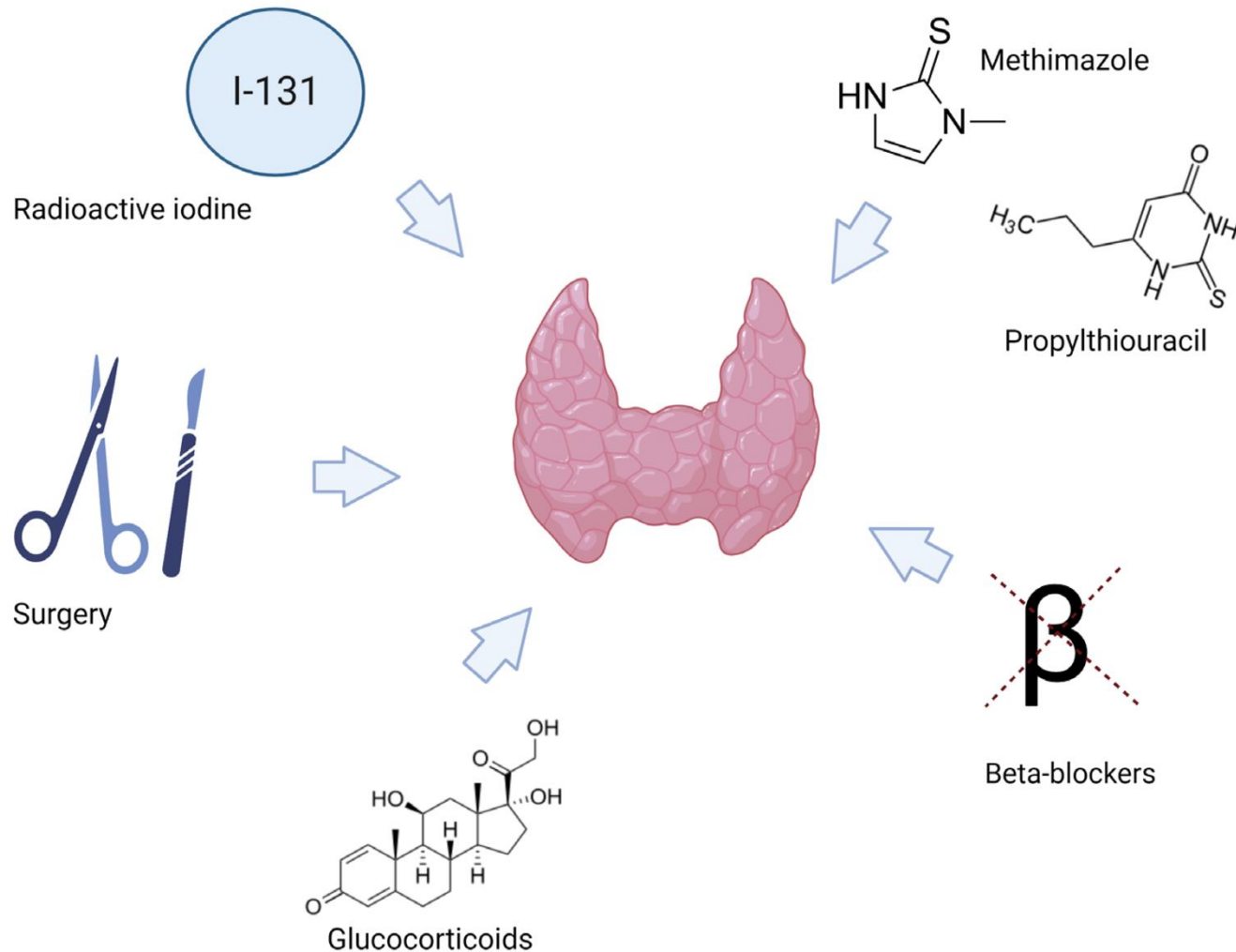
Amiodarone

Impaired $T_4 \rightarrow T_3$

Mechanisms uncertain or multifactorial

Estrogen pregnancy
Ethionamide
Tyrosine kinase inhibitors (imatinib, sunitinib)
Lovastatin, simvastatin

Treatment of hyperthyroidism



DRUGS USED FOR TREATMENT OF HYPERTHYROIDISM

1. Hormone synthesis inhibitors (Antithyroid drugs) : thionamide

- Propylthiouracil (PTU) , Carbimazole, Methimazole

2. Inhibit Iodic trapping (Iodic Inhibitors):>>not use in practical

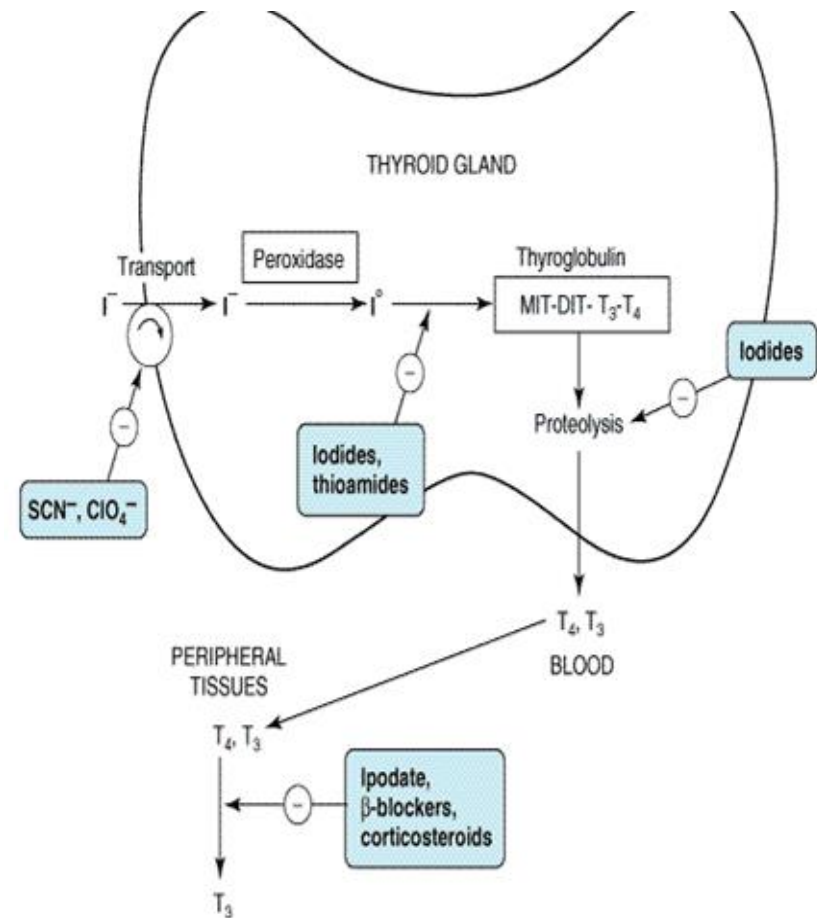
- Perchlorates (ClO_4^-), Thiocyanates (SCN^-), Nitrates (NO_3^-)

3. Hormone release inhibitors:

- High concentration of Iodine, Iodides

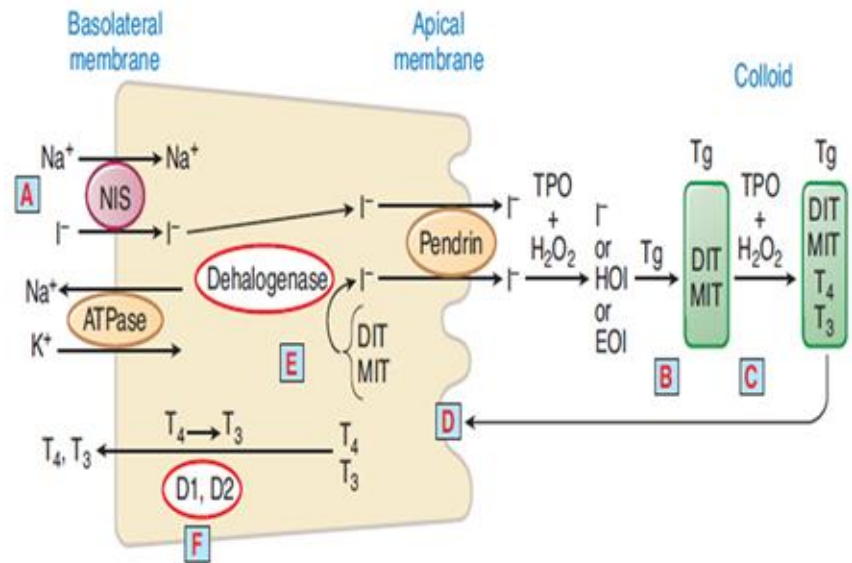
4. Destroy thyroid tissue:

- Radioactive iodine (^{131}I)



1. Antithyroid drugs

- **Thionamides group**
- **Propylthiouracil (PTU), Methimazole (thiamazole or MMI), Carbimazole**
- Mechanism of action: Inhibit hormone synthesis
- 1. Interfere oxidation and iodination processes
- 2. inhibiting peroxidase enzyme
- PTU also inhibits peripheral deiodination of $T_4 \rightarrow T_3$
- Methimazole: more potent and longer acting than PTU



PTU, MMI →

PTU →

KEY		
	METABOLIC STEP	INHIBITOR
A	Iodine transport	ClO ₄ ⁻ , SCN ⁻
B	Iodination	PTU, MMI
C	Coupling	PTU, MMI
D	Colloid Resorption	Colchicine, Li ⁺ , I ⁻
E	Deiodination of DIT + MIT	Dinitrotyrosine
F	Deiodination of T ₄	PTU

Pharmacokinetics

Well absorbed orally, widely distributed

Preparations: PTU: 50 mg tab, Methimazole: 5 & 10 mg tab and Carbimazole: 2.5/5 mg tab

Carbimazole is converted to its active metabolite, methimazole

PTU – highly plasma protein bound

Partly metabolized in the liver and the thyroid gland

excreted in the urine unchanged

Selected Pharmacokinetic features of antithyroid drugs

	PTU	Methimazole
Plasma protein binding	75%	not
Plasma $T_{1/2}$	75 min	4-6 hr
V_d	20 L	40 L
Concentrated in thyroid	yes	yes
Dose frequency	1-4 times	One or twice /day
Transplacental passage	low	low
Level in breast milk	low	low

ADVERSE EFFECTS

- **Common adverse effects:** **maculopapular rash**, GI side effects, arthralgia (treat with antihistamine drugs)
- Hypothyroidism (overtreatment)
- Rare: dermatitis, antineutrophilic cytoplasmic antibody (ANCA) associated vasculitis, lupus-like reaction
- Hepatitis: PTU
- Cholestatic dysfunction: methimazole
- Agranulocytosis (reversible) – most serious reactions
 - PTU: idiosyncrasy
 - Methimazole: dose-related

Therapeutic uses

1) Non-operative therapy of hyperthyroidism, **Graves' disease**

2) Preoperative therapy of hyperthyroidism: combined with iodide

3) Thyrotoxicosis crisis: combined with propranolol, larger dose of iodide

HYPERTHYROIDISM IN PREGNANCY

- Occurs about 0.2 – 0.4% of all pregnancies
- Due to Grave's Disease (common), Toxic nodules, Thyroiditis
- Medical therapy – **TREATMENT OF CHOICE**
PTU during the 1st trimester due to less teratogenicity

- then switched over to Methimazole due to its lower hepatotoxic potential
- methimazole during the first trimester of pregnancy: risk of birth defects (**MMI embryopathy**)
 - minor dysmorphic features, choanal atresia and/or esophageal atresia, growth retardation, and developmental delay

2. Iodide solution

High dose of iodide (1.5-150 mg) or >6 mg/d: treatment of hyperthyroidism

↓ thyroid hormone release (rapid effect)

inhibit peripheral conversion of T4 into T3

Preparations:

Lugol's solution (aqueous iodine oral solution, strong iodine solution):

5% iodine + 10% potassium iodide

Saturated solution of potassium iodide (SSKI): yields a dose of 38 mg of iodine/dose

Rapid response (within 24 hr.) but high rate of relapse

Cautions:

“escape phenomenon”: after use for 10-15 days, the effects disappear!! → use only short duration!

Increase intraglandular store of iodine → decrease onset effect of thioamide

ADVERSE REACTIONS

1) Acute effects:
hypersensitivity to iodine

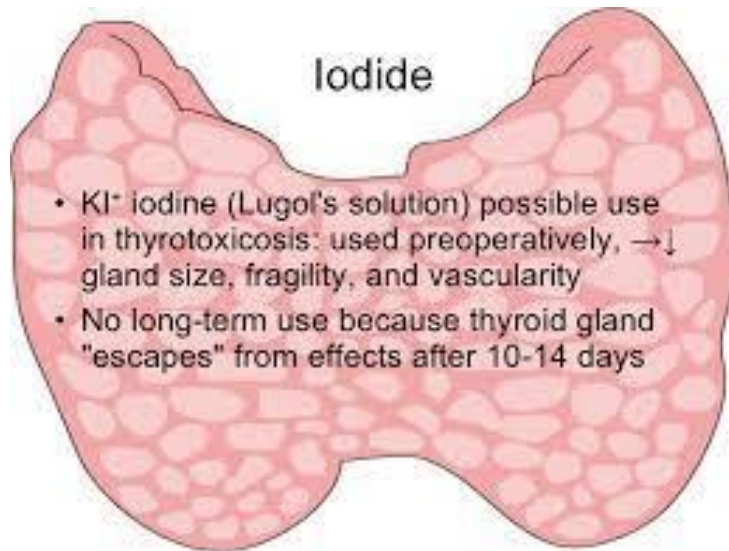
swelling of lips, eyelids,
angioedema of larynx,
fever, joint pain, petechial
hemorrhages

**(2) Chronic intoxication
(iodism):** dose-dependent

- Unpleasant taste, burning in mouth, soreness of teeth and gum, salivation, skin lesion
- Should not used in pregnancy

Long term use of high
doses – Hypothyroidism
and goiter

Therapeutic use of iodide



1) Preoperative therapy of hyperthyroidism

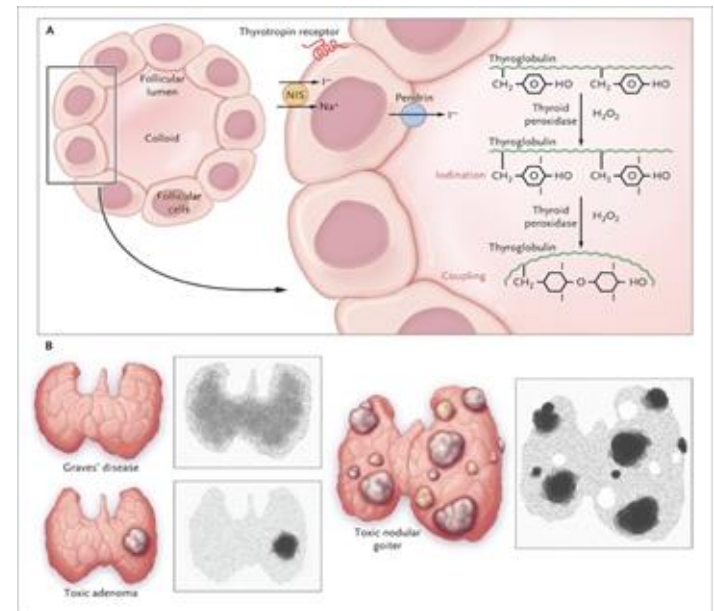
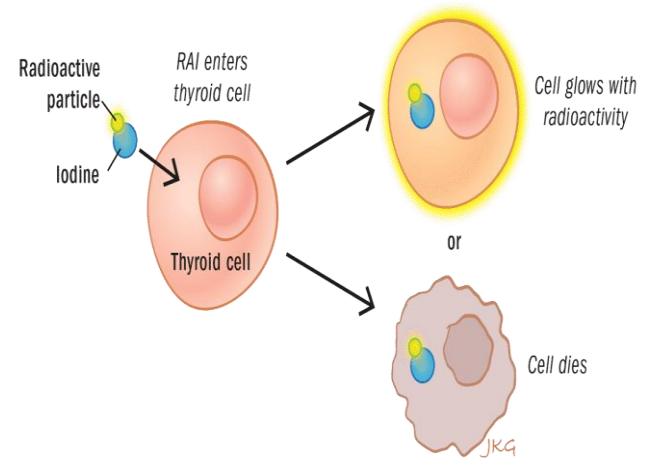
2) Thyrotoxicosis crisis: combined with PTU

3) Prophylaxis of endemic goiter

potassium iodide (KI) - limiting the potential damage to thyroid gland by radiation emergencies (30-100 mg iodide, decrease uptake of radioactive iodine)

3. Radioactive Iodine

- I^{131} : only isotope used in treatment of thyrotoxicosis (destroy thyroid gland)
 - I^{123} : for diagnostic to measure 24 hr. iodine uptake and thyroid image, emit γ ray with $T_{1/2}$ 13 hr.
 - Swallowed I^{131} trapped and concentrated in thyroid follicles
 - Beta radiation 90 % - penetration upto 0.5 mm
 - Gamma radiation - deep penetration
 - Radioactive half-life 8 days
 - Maximal effects –3 months
 - Average therapeutic dose: 4-15 mCi
- Uses: Overactive or enlarged thyroid, thyroid cancer



Radioactive (I^{131})

INDICATIONS

- **Most common indications:**
hyperthyroidism due to Graves' disease and Toxic Nodular Goiter
- Indicated in elderly patients
- allergy to PTU
- recurrent hyperthyroidism
- patients with systemic diseases contraindicating of surgery

Advantages :

- Simple, inexpensive
- No surgical risk, scar or injury to parathyroids and nerves
- Control of hyperthyroidism is permanent

The main indications for RAI therapy include the following conditions

1. Hyperthyroidism due to:
 - a. Grave's disease
 - b. Toxic multinodular goitre or
 - c. Hyperfunctioning thyroid nodules
2. Non-toxic multinodular goitre
3. Thyroid cancer.

Contra-indications for RAI therapy

1. Pregnancy
 2. Breast feeding
 3. Severe uncontrolled thyrotoxicosis
-

Disadvantages and Adverse effects of radioactive (I^{131})

- Permanent hypothyroidism
- Cannot be used during pregnancy/breast feeding
- Not suitable for young patients
- Focal soreness in the neck
- Damage to fetal thyroid
- Thyroid carcinoma, Leukemia
- Radiation induced genetic damage

ตารางที่ 1 เปรียบเทียบการรักษาไทรอยด์เป็นพิษด้วยยาต้านไทรอยด์ การผ่าตัด และไอโอดีนรังสี

	ยาต้านไทรอยด์	การผ่าตัด	ไอโอดีนรังสี (^{131}I)
ข้อบ่งชี้	- เด็ก, วัยรุ่น - ตั้งครรภ์ - Hyperthyroidism ที่เพิ่งได้รับการวินิจฉัยและมีอาการไม่มาก	- ต่อมไทรอยด์มีขนาดใหญ่ส่งผลให้เกิด compressive symptoms เช่น หายใจลำบาก หรือกลืนลำบาก - สงสัยมะเร็งไทรอยด์ร่วมด้วย หรือ มี cold nodule	- มีการกลับเป็นซ้ำหลังการผ่าตัดหรือหยุดยาต้านไทรอยด์ (relapse) - ยาต้านไทรอยด์ไม่สามารถคุมอาการได้ (failed medication) - ผู้ป่วยมีผลข้างเคียงรุนแรงจากการใช้ยาต้านไทรอยด์
ข้อดี	- ควบคุมอาการได้เร็ว - สามารถใช้ได้ในผู้ป่วยตั้งครรภ์	โอกาสหายขาดสูง	- โอกาสหายขาดสูง - ราคาไม่แพง
ข้อเสีย	- โอกาส relapse สูง - อาจพบผลข้างเคียง (side effects) เช่น fever, rash, pruritus, hepatitis, glomerulonephritis	ผลแทรกซ้อนจากการผ่าตัด เช่น recurrent laryngeal nerve injury, post-surgical hypothyroidism, hypoparathyroidism	- มีข้อห้ามในผู้ป่วยตั้งครรภ์ และหญิงให้นมบุตร - เห็นผลการรักษาค่อนข้างช้า

Adjuvant therapy of hyperthyroidism

● 1. **β- blockers**

- Antagonize sympathetic effect of thyrotoxicosis
- provides quick relief for tremor, palpitation, tachycardia
- propranolol 20-80mg/ atenolol 25-200 mg
- Propranolol high dose: decrease peripheral conversion of T4 to T3

● 2. **Calcium channel blockers**: diltiazem 60-120 mg

● 3. **Dexamethasone** (0.5-1 mg) 2-4 times daily: inhibit peripheral conversion of T4 to T3

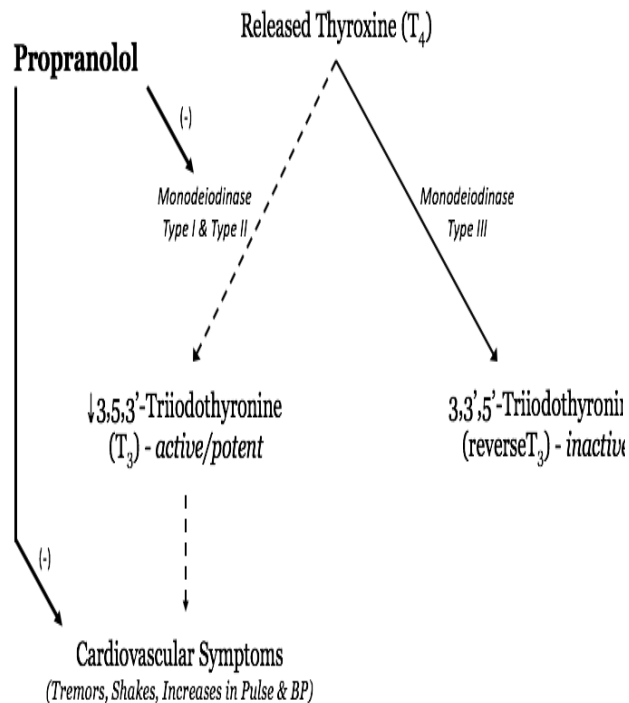
● 4. **Lithium carbonate**: inhibit thyroid release

- Use in patients with problem of iodine allergy

Beta-adrenergic receptor blockade in the treatment of thyrotoxicosis

ADRs: Fatigue, sexual dysfunction, Sinus bradycardia, heart block

**Special considerations:
Use with caution in asthma, DM, CHF, depression**



ยา	ขนาดยา (มิลลิกรัม)	การบริหารยา (ครั้ง/วัน)	ข้อควรพิจารณา
Propranolol	10-40	3-4	- Nonselective beta-adrenergic blockade - มีประสบการณ์การใช้ยาวนานที่สุด - ในขนาดสูง อาจช่วยยับยั้งการเปลี่ยนจาก T₄ เป็น T₃ ได้ - เป็นทางเลือกสำหรับหญิงตั้งครรภ์และให้นมบุตร
Atenolol	25-100	1-2	- Relative beta-1 selectivity - ไม่แนะนำให้ในหญิงตั้งครรภ์
Metoprolol	25-50	2-3	- Relative beta-1 selectivity - มีแบบให้วันละครั้ง (extended release)



คำแนะนำ
สำหรับภาวะความผิดปกติของต่อมไทรอยด์
พ.ศ. ๒๕๖๘

จัดทำโดย
สมาคมต่อมไร้ท่อแห่งประเทศไทย

Thank You



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